

THE VOLUME OF WHITE MATTER IN A PERSON'S BRAIN PEAKS AROUND AGE 40



MORALITY



FUTURE

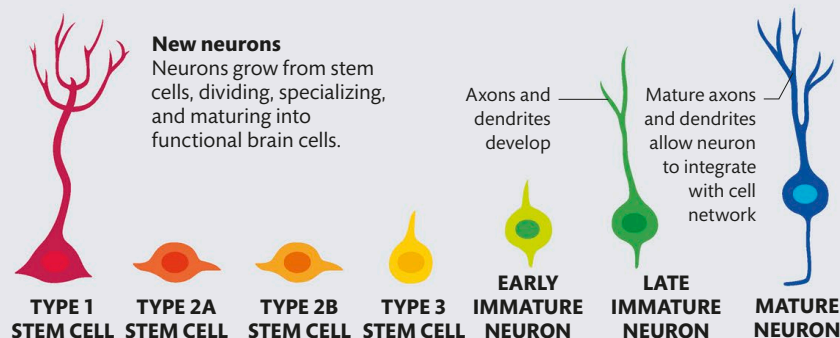


WORK



Neurogenesis

Neurogenesis is the development of new neurons by neural stem cells (cells that can become other cells). In a range of mammals, neurogenesis happens in the hippocampus and olfactory areas and continues throughout life, with new neurons being produced regularly. The same is thought to be true in humans, although the evidence is mixed. Neurogenesis may also play a role in learning and memory.



Disrupting memories

New brain cells help store information, so boosting neurogenesis in the brain can improve learning into adulthood. However, it also has a role to play in forgetting. Adding in new brain cells with new connections disrupts existing memory circuits by competing with them. This means there is an optimal level of neurogenesis, which balances learning ability with retaining older memories.

Memory storage

Due to the creation of new brain cells, hippocampal memories may degrade before they can be stored in the cortex. This might explain why we are unable to remember our infancy.

